

Design of a Dual-Band Polarization-Insensitive Rectifying Metasurface for Wireless Power Transfer

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Abstract — This paper proposes a dual-band polarization-insensitive rectifying metasurface for wireless power transfer (WPT). The unit cell comprises two hollowed dipoles, with an air substrate attached to the reflector plane. Two Schottky diodes are integrated into each unit cell to realize polarization insensitivity. By eliminating the matching network, the design minimizes losses and maximizes rectifying efficiency, with the Schottky diode directly conjugately matched to the metasurface structure. The unit has a compact dimensions of $0.21\lambda_0 \times 0.21\lambda_0 \times 0.05\lambda_0$. An array of 4×4 units is proposed, and a series-parallel dc network is applied to increase and stabilize the output voltage. The proposed metasurface is designed to operate at 2.4 GHz and 5.8 GHz, with simulation results showing that the total efficiency reaches 61.9% and 46.6%, respectively, at an input power of 10 dBm. Measurements show that the peak efficiency reaches 59.0% at 2.41 GHz and 40.1% at 5.33 GHz, respectively. The proposed rectifying metasurface demonstrates polarization insensitivity, making it ideal for wireless power transfer applications.

Keywords — wireless power transfer, rectifying metasurface, dual-band, polarization insensitivity.

I. INTRODUCTION

As industrialization progresses, the proliferation of sensors has introduced significant challenges in providing power to these devices. Over recent decades, microwave wireless power transfer (MWPT) technology has gained popularity for its convenience in wirelessly powering various terminals [1]. Rectennas are traditionally employed to receive electromagnetic power and convert it into DC power. Compared with traditional single-band rectenna, dual-band rectennas offer enhanced flexibility, enabling power reception across multiple frequencies [2]. However, traditional rectennas faces limitations in radiation efficiency, which in turn restricts rectifying efficiency. Besides, conventional rectennas employ matching networks to align the antenna and diode impedance to 50Ω , introducing additional losses [3]–[5].

Metamaterials, consisting of sub-wavelength units in a periodic arrangement, have been studied extensively. Metasurfaces are 2-D metamaterials with planar structures. By adjusting the arrangement of unit cells flexibly, metasurfaces can perform various functions, such as perfect absorber [6], frequency-selective surface [7], reflectarray [8], etc. Metasurfaces as perfect absorbers provide a novel solution for WPT receiver design. When integrated with rectifiers, rectifying metasurfaces enhance the capture of electromagnetic energy, leading to higher conversion efficiency.

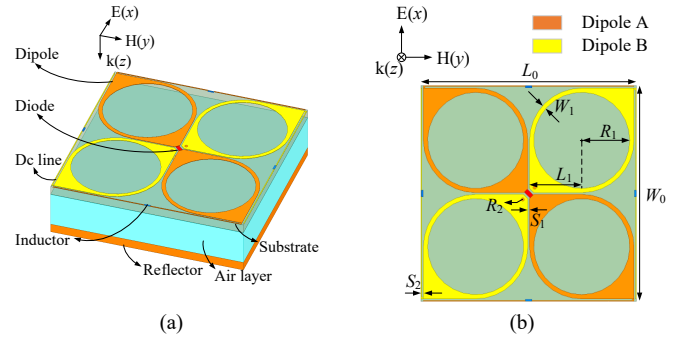


Fig. 1. Structure of a unit cell. (a) 3D view. (b) Top view. $L_0 = 25.6$, $W_0 = 25$, $L_1 = 6.2$, $W_1 = 0.5$, $R_1 = 5.7$, $R_2 = 0.2$, $S_1 = 0.2$, $S_2 = 0.3$. (Unit: mm)

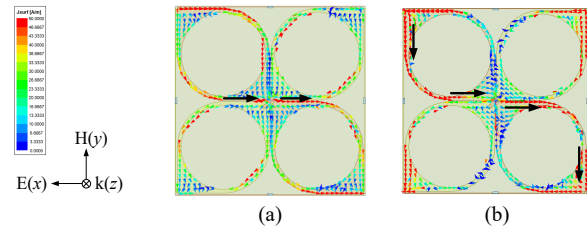


Fig. 2. Surface current density distribution at (a) 2.4GHz and (b) 5.8GHz, respectively.

Rectifying metasurfaces are typically characterized by narrow bandwidths due to the closely coupled unit cells. Multi-band rectifying metasurfaces have been extensively studied to harvest more electromagnetic power across multiple frequencies, thereby achieving efficient rectifying while maintaining structure simplicity [9]–[12].

This paper introduces a design approach for a dual-band polarization-insensitive rectifying metasurface. Considering the large real part of rectifying diode's impedance, two dipoles are employed as harvesters in unit cells. These dipoles are spindle-shaped to facilitate multiple current paths, realizing dual-band characteristics. The dipoles are arranged orthogonally, with each dipole connected to a Schottky diode to adapt to different polarization angles. Hollowed structures are elaborately designed to tune the impedance for matching with free space impedance and conjugate matching with the rectifying diodes. Furthermore, a series-parallel DC network is implemented to increase and stabilize the output voltage. Measurements validate the design and demonstrate the potential of the proposed rectifying metasurface.

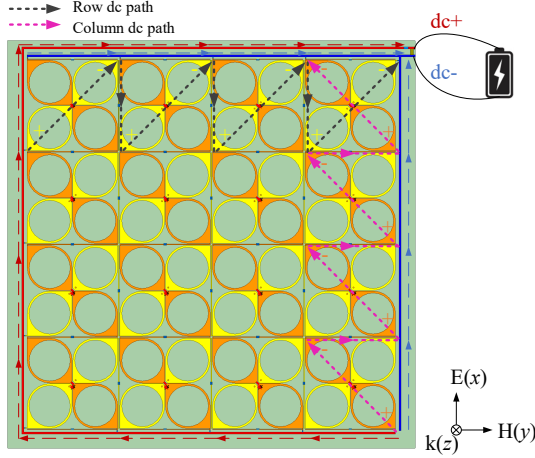


Fig. 3. Array of proposed rectifying metasurface.

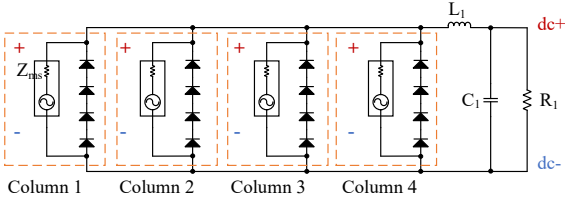


Fig. 4. Equivalent circuit schematic of the rectifying metasurface.

II. RECTIFYING METASURFACE DESIGN

The unit cell structure of the proposed metasurface is shown in Fig. 1. It consists of two dipoles, a substrate, an air layer, and a metallic reflector. The dipoles are implemented on a 1 mm thick substrate of F4B with permittivity of 2.65 and loss tangent of 0.002. The two dipoles are rotationally symmetric and are placed orthogonally to achieve polarization insensitivity. For analytical convenience, the dipoles are denoted as dipole A and dipole B. One Schottky diode HSMS286C is integrated at the center of the top layer and connected to dipole A, and the other at the center of the bottom layer, connected to dipole B through vias. A 4 mm thick air layer attaches the substrate to a metallic reflector. The proposed unit has a compact size of $0.21\lambda_0 \times 0.21\lambda_0 \times 0.05\lambda_0$.

The diodes usually present a high impedance, thus dipoles which have high impedances are chosen as the unit cell to capture electromagnetic power. The dipole is designed to be a spindle-shaped structure with multiple edges to couple with adjacent ones. As a results, the proposed structure resonates at 2.4 GHz and 5.8 GHz, respectively. To conjugately match the diodes impedance at these resonances, hollowed geometries are used to enhance impedance tuning. Fig. 2 illustrates the surface current density distribution of the proposed metasurface. It is obvious that the current mainly concentrates along the central parts of structure at 2.4 GHz, while at 5.8 GHz the current density shrinks to only half dipole. This demonstrates that two modes are excited at the two different frequencies. Notably, by adjusting the parameters of the structure, it's possible to make the metasurface resonate at different frequencies.

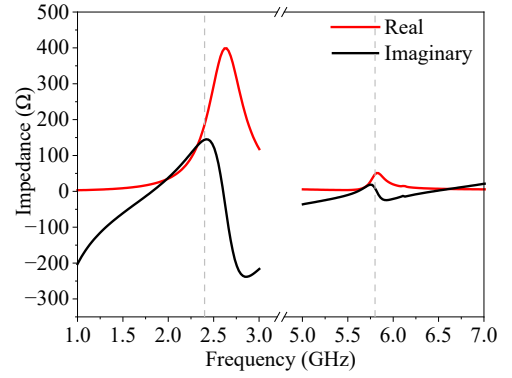


Fig. 5. Simulated impedance of metasurface

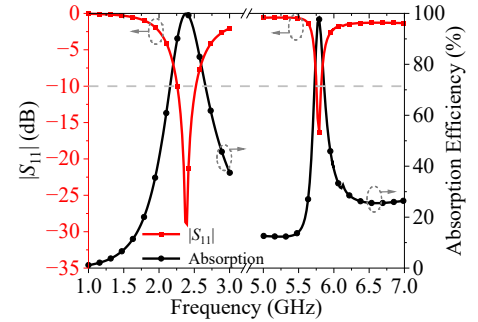


Fig. 6. Simulated S parameter and absorption efficiency

The array of a 4×4 rectifying metasurface is illustrated in Fig. 3. Considering the low voltage generated by a single diode, a series-parallel network is adopted to increase the output voltage. In each column/row, 4 dipoles are connected in series and the resulting outputs are connected in parallel. The circuit network topology of the rectifying network is shown in Fig. 4, where the electromagnetic power captured by the metasurface is equivalent as the source, and the impedance looking into the metasurface from the diode is equivalent as the impedance Z_{ms} . The circuit of dipole B is similar to that.

III. RESULTS AND ANALYSIS

The RF-DC efficiency of rectifying metasurface η_{RF-DC} can be expressed with a product of two efficiencies,

$$\eta_{RF-DC} = \eta_h \times \eta_{ac-DC} \quad (1)$$

where η_h is the harvesting efficiency of the metasurface, and η_{ac-DC} is the rectifying efficiency of the diode. System efficiency requires both efficiencies to be maximized, which indicates that the metasurface should be a perfect absorber and the diode should be exactly conjugately matched with the metasurface.

Due to the nonlinear characteristic of diodes, their impedances vary as a function of frequency, power and load impedance. To obtain high rectifying efficiency at both frequencies, a load impedance 270Ω is selected after extensive trade-off analysis, and input power is set at 10 dBm. The diode impedance calculated by full-wave simulation is $(166-j159) \Omega$ at 2.4 GHz and $(42-j55) \Omega$ at 5.8 GHz. By carefully tuning the parameters of the metasurface, conjugate impedance is achieved, as shown in Fig. 5. The impedance of

metasurface is $(186+j143) \Omega$ at 2.4 GHz and $(45+j12) \Omega$ at 5.8 GHz. Consequently, the impedance of the metasurface and diode are almost matched conjugately, resulting in small losses.

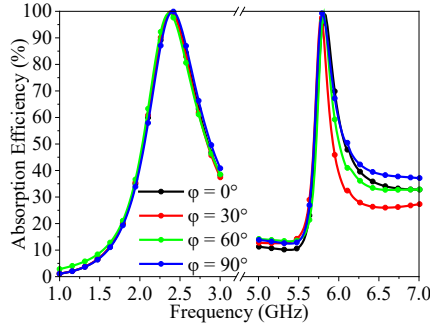


Fig. 7. Simulated absorption efficiency for various polarization angles.

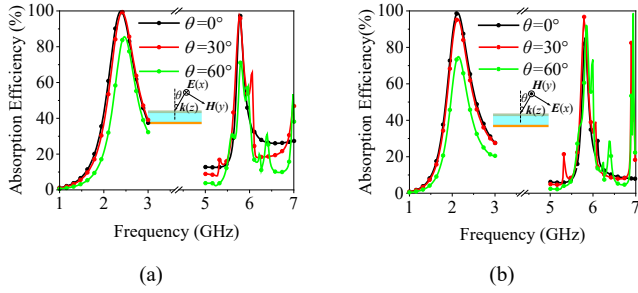


Fig. 8. Simulated absorption efficiency for various incident angles with (a) TE polarization and (b) TM polarization.

The electromagnetic power harvesting rate of the metasurface can be calculated as $\eta_h = 1 - |S_{11}|_{xx}^2 - |S_{11}|_{xy}^2$. The full-wave simulation results of harvesting efficiency are shown in Fig. 6, and it reaches 99% at 2.45 GHz and 97% at 5.8 GHz, respectively.

The proposed metasurface maintains a high harvesting efficiency for different polarization angles. As depicted in Fig. 7, absorption efficiency remains over 95% at two frequencies for all polarization angles. Fig. 8 illustrates the harvesting efficiency with respect to the incident angle. At 2.4 GHz, the harvesting efficiency over 80% is achieved for incident angle up to 60°. At 5.8 GHz, the harvesting efficiency exceeds 70% as the incident angle rises to 60°.

The impedance of metasurface is co-simulated with the diode, and total efficiency can be calculated. The average total efficiency can be expressed as the sum of two total efficiencies at two frequency points,

$$\eta_{total} = (\eta_{total@2.4GHz} + \eta_{total@5.8GHz})/2 \quad (2)$$

Since the absorption efficiencies of two frequencies are already known, bringing formula (1), the total efficiency can be written as,

$$\begin{aligned} \eta_{total} &= \eta_{h@2.4GHz} \times \eta_{ac-DC@2.4GHz}/2 \\ &\quad + \eta_{h@5.8GHz} \times \eta_{ac-DC@5.8GHz}/2 \\ &\approx (0.99 \times \eta_{ac-DC@2.4GHz} + 0.97 \times \eta_{ac-DC@5.8GHz}) \end{aligned} \quad (3)$$

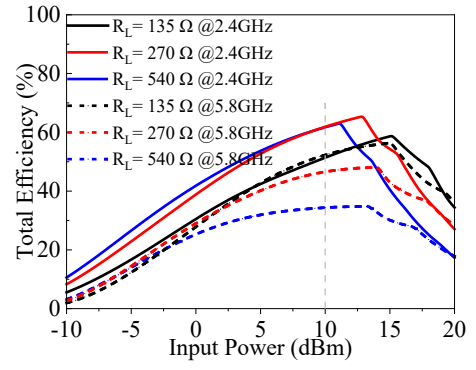


Fig. 9. Simulated total efficiency with various load impedance at 2.4 GHz and 5.8 GHz.

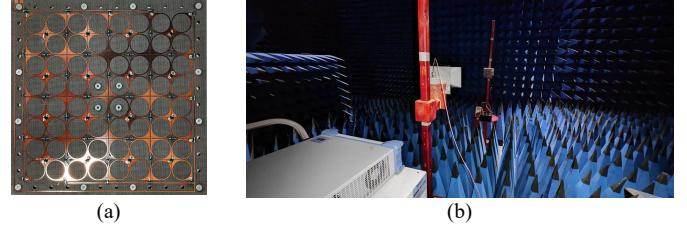


Fig. 10. Photograph of (a) proposed metasurface, and (b) experiments setup.

Table. 1. Measured and simulated total efficiencies.

Simulated η_{max}	61.9 @ 2.4 GHz	46.6 @ 5.8GHz
Measured η_{max}	59.0 @ 2.41 GHz	40.1 @ 5.33GHz

With determined frequency and input power, the rectifying efficiency relies on load impedance. Thus, load impedance is optimized to maximize the average total efficiency. The simulated total efficiency across different load impedances at 2.4 GHz and 5.8 GHz is depicted in Fig. 9. The highest efficiency occurs at an impedance of 540 Ω at 2.4 GHz and 135 Ω at 5.8 GHz. A compromise load impedance of 270 Ω is selected for practical implementation.

To validate the design, a prototype is fabricated and measured. The manufactured sample is depicted in Fig. 10. (a) with an overall size of 120 mm $\times$ 120 mm. Experiments setup is depicted in Fig. 10. (b), a horn antenna is employed as transmitting antenna. Table. 1 compares simulated and measured total efficiencies. For lower band, the peak total efficiency reaches 59.0% at 2.41GHz, and for higher band, the peak total efficiency reaches 40.1% at 5.33GHz. While the simulated total efficiency reaches 61.9% and 46.6% at 2.4 GHz and 5.8 GHz, respectively. The measured peak efficiencies are in good agreement with simulation, but the peak efficiency point in higher band is shifted to 5.33 GHz.

The polarization robustness of the design is further confirmed in Fig. 11. The efficiency fluctuations remain below 4.8% for lower band and 6.2% for higher band, which demonstrates the good polarization-insensitive characteristic of proposed metasurface.

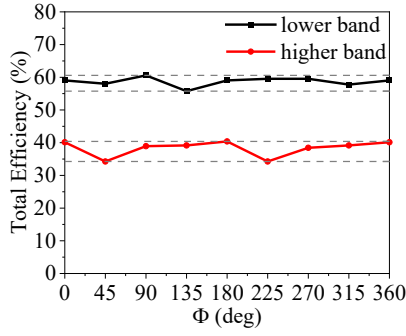


Fig. 11. Measured total efficiency at different polarization angle.

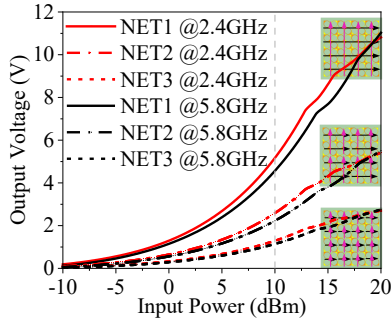


Fig. 12. Output voltage of different networks.

The proposed rectifying metasurface is capable of rectifying at dual frequencies. However, the voltage generated by the diode is low. If a conventional DC network is applied, the output voltage of $N \times N$ units are connected in parallel, leading to a low output voltage, and load impedance becomes $1/N^2$, making it difficult to drive the load. Thus, a series-parallel dc network is utilized, as seen in Fig. 3. Four columns and four rows split the units in the 4×4 array. Every unit in a column and row is connected in series, helping increase the output DC voltage, and the DC output of each column and row is connected in parallel, respectively. The DC network provides flexibility of conveniently adjusting output DC voltage. The output DC voltages of three different topologies are shown in Fig. 12. In network 1 and 2, units are divided into 4 and 8 groups, respectively, where 4 and 2 units are connected in series in one group, and these 4 and 8 groups are connected in parallel, respectively. In network 3, every element output is connected in parallel. For the proposed network, i.e., network 1, the output DC voltage in the input power of 10 dBm is 5.2 V at 2.4 GHz and 4.6 V at 5.8 GHz, respectively. The output DC voltage is 2.6 V and 1.3 V at 2.4 GHz and 2.2 V and 1.1 V at 5.8 GHz for network 2 and 3, respectively. Within the range of 4.5 V – 5.5 V, the proposed DC network is ideally suitable for loads driving, and other voltage requirements can be met by flexibly changing the topology of DC networks.

IV. CONCLUSION

This paper proposes a dual-band polarization-insensitive rectifying metasurface operating at 2.4 GHz and 5.8 GHz. The

metasurface consists of hollowed dipoles that harvest electromagnetic energy, with Schottky diodes directly integrated into the structure without impedance matching networks. Simulation results demonstrate peak total efficiencies of 61.9% at 2.4GHz and 46.6% at 5.8GHz, with an input power of 10dBm. Experimental measurements corroborate these findings with measured peak efficiencies of 59.0% at 2.41GHz and 40.1% at 5.33GHz. Additionally, the metasurface exhibits polarization insensitivity, making it a promising candidate for practical wireless power transfer systems. Besides, a series-parallel DC network is employed to increase and stabilize output voltage, further enhancing the versatility of the proposed design.

ACKNOWLEDGMENT

This work was supported in part by National Natural Science Foundation of China (NSFC) under Grant U22A2015 and in part by Sichuan Science and Technology Program under Grant 2024YFHZ0282. (Corresponding author: Liping Yan and Changjun Liu)

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